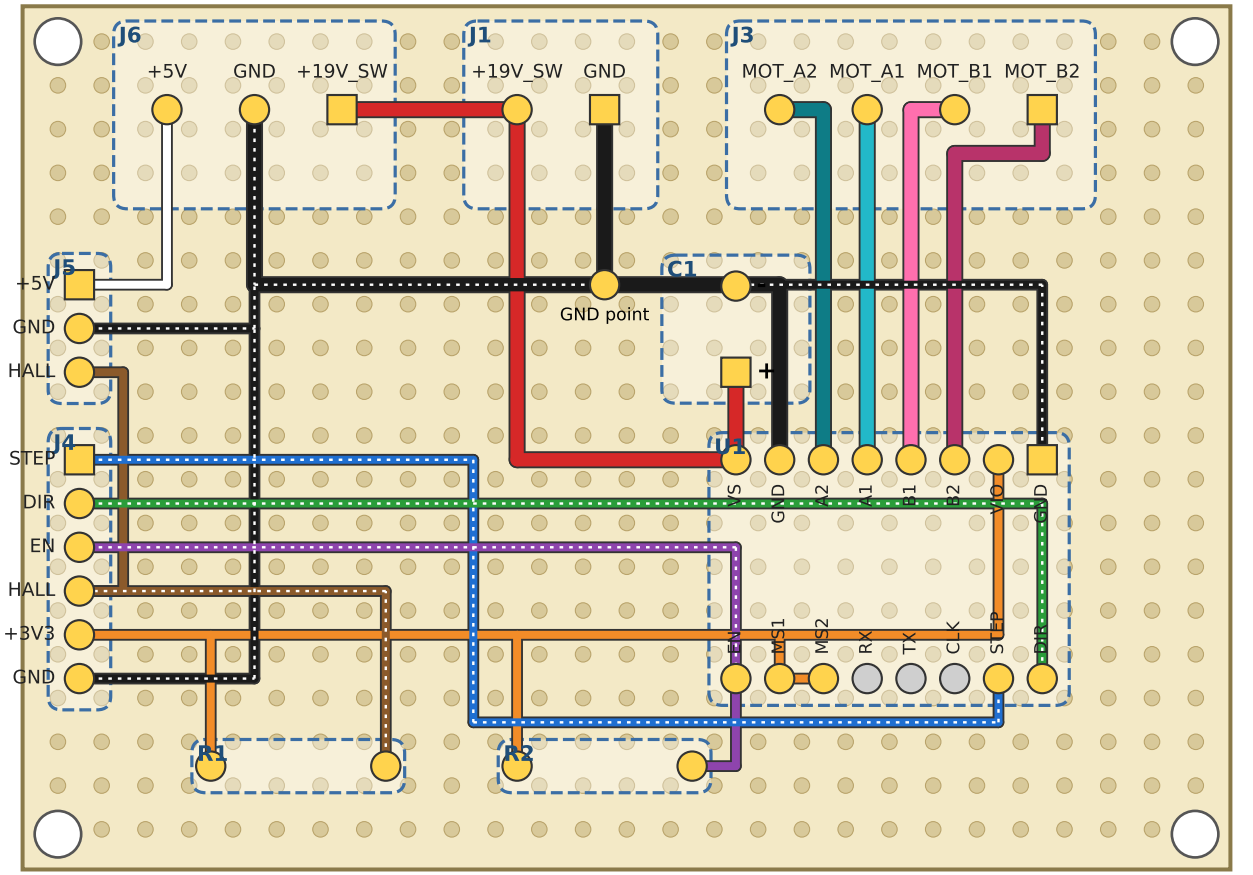
Piper-Watch Level 2 perfboard (70 x 50)



TOP VIEW - back of the case at the top.

Wires are soldered on the UNDERSIDE, but drawn as seen from the top.

Thick = heavy wire (19 V, GND, motor).

Dotted stripe = INSULATED wire (it crosses another run); the rest can be bare.

Grey pads are not connected. TMC2209 square pad = pin 1 (GND).

J6 / J1 / J3: the screw openings face the back edge.

C1 100uF 35V (at U1 VS)

J1 19V in: + - left to right

J3 motor: A2 A1 B1 B2 (left to right on the board)

J4 to ESP32: STEP DIR EN HALL 3V3 GND

J5 to Hall: 5V GND OUT

J6 to buck: 19V, GND, 5V

R1 10k (Hall pull-up)

R2 10k (opt., EN pull-up)

U1 BTT TMC2209

19 V (switched)

GND

5 V (Hall)

3V3 (from the ESP32)

STEP

DIR

EN

HALL

coil A2

coil A1

coil B1

coil B2